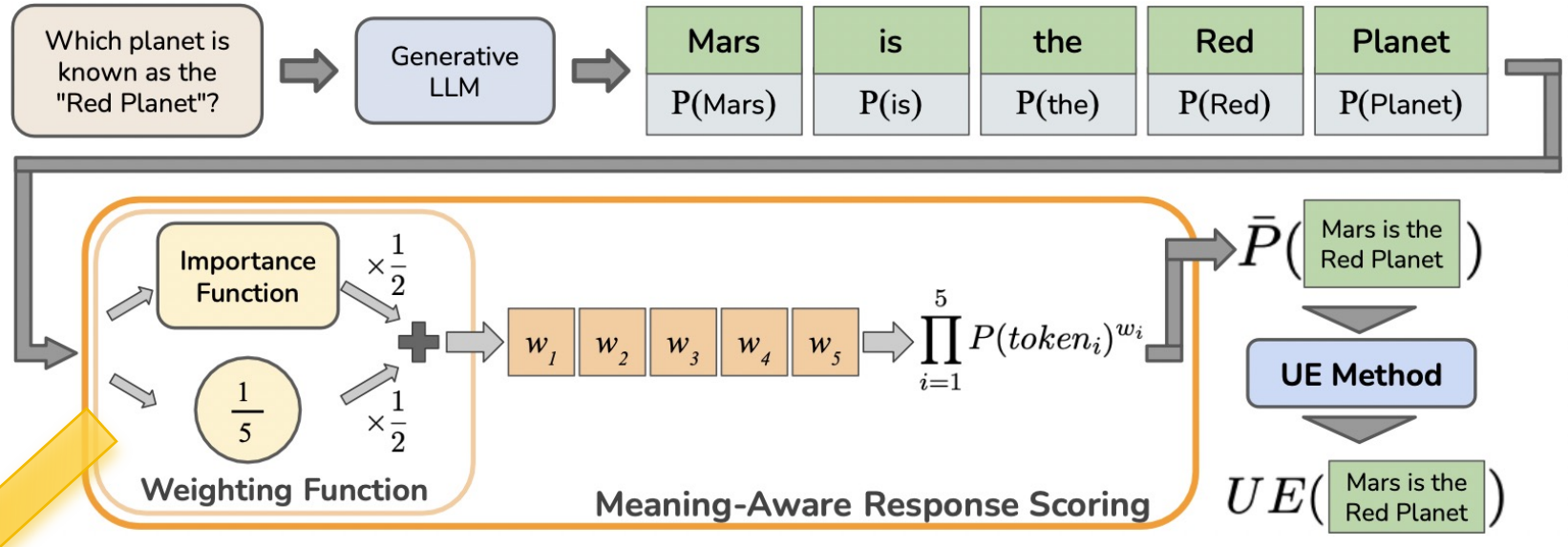
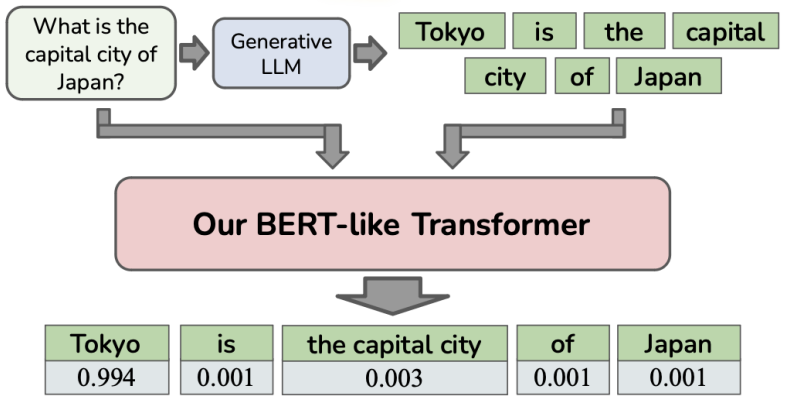
MARS

MARS[6]

change token-level probabilities into importance weights reflecting how much each token contributes to meaning.





$$w(\mathbf{s}, \mathbf{x}, L, l) \triangleq \frac{1}{2L} + \frac{u(\mathbf{s}, \mathbf{x}, l)}{2}$$ ✅

each word's contribution to the sentence's meaning in the question context might vary

$$\tilde{P}(\mathbf{s}|\mathbf{x}, \theta) = \prod_{l=1}^{L} P(s_l|s_{<l}, \mathbf{x}; \theta)^{\frac{1}{L}}$$ ❌

[6] MARS: Meaning-Aware Response Scoring for Uncertainty Estimation in Generative LLMs, ACL 2024

MARS[6]

- **Task Type:** Open-ended generation

- **Introduction:** MARS reflects each token's semantic contribution to a generated answer, which replaces length-normalized scoring by weighting token probabilities accordingly. It quantifies prediction uncertainty and offers a meaning-aware scoring method for token-level uncertainty estimation in generative LLMs.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Token-level	White-box	Post-hoc	Medium	Ranking

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